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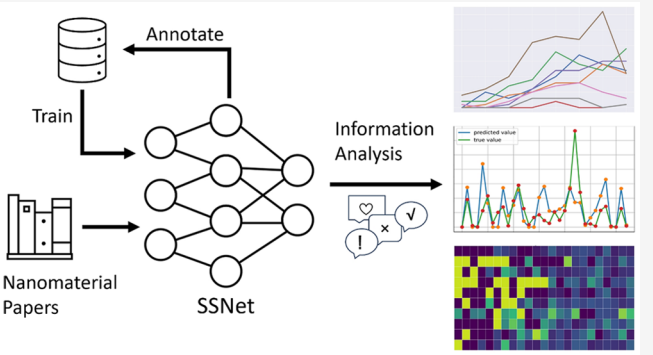
Opinion Mining by Convolutional Neural Networks for Maximizing Discoverability of Nanomaterials

Tong Xie,*,¶ Yuwei Wan,¶ Haoran Wang, Ina Østrøm, Shaozhou Wang, Mingrui He, Rong Deng, Xinyuan Wu, Clara Grazian, Chunyu Kit,* and Bram Hoex*

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ABSTRACT: The scientific literature contains valuable information that can be used for future applications, but manual analysis presents challenges due to its size and disciplinary boundaries. The prevailing solution involves natural language processing (NLP) techniques such as information retrieval. Nonetheless, existing automated systems primarily provide either statistically based shallow information or deep information without traceability, thereby falling short of delivering high-quality and reliable insights. To address this, we propose an innovative approach of leveraging sentiment information embedded within the literature to track the opinions toward materials. In this study, we integrated material knowledge into text representation and constructed opinion data sets to hierarchically train deep learning models, named as Scientific Sentiment Network (SSNet). SSNet can effectively extract knowledge from the energy material literature and accurately categorize expert opinions into challenges and opportunities (94% and 92% accuracy, respectively). By incorporating sentiment features determined by SSNet, we can predict the ranking of emerging thermoelectric materials with a 70% correlation to experimental outcomes. Furthermore, our model achieves a commendable 68% accuracy in predicting suitable nanomaterials for atomic layer deposition (ALD) over time. These promising results offer a practical framework to extract and synthesize knowledge from the scientific literature, thereby accelerating research in the field of nanomaterials.



1. INTRODUCTION

The role of the scientific literature is to communicate recent advances and keep researchers informed so they can continue to make breakthroughs and advance their field. The increasing number of publications makes manually selecting and reading relevant papers time-consuming and difficult,[1−4] which calls for more efficient ways to extract up-to-date knowledge and inspire researchers. Several studies have explored the automated retrieval of information from scientific literature using natural language processing (NLP) techniques.[5−8] In materials science, significant effort has been devoted to extracting material mentions[3,9] or synthesis recipes.[10−12] These applications are based on named-entity recognition (NER), linking material names to their co-occurring entities, which help answer large-scale questions, such as "What are the most frequent methods for TiO_2 synthesis?"[9]

In addition to shallow information, researchers also need high-level information to make predictions over thousands of materials. Nevertheless, there has been less effort to develop tools that extract high-level information from materials science texts.[9] Thus, it will be of great benefit if the published literature can be integrated and summarized to provide relevant information about the properties or performance of materials. Recent attempts to predict suitable materials for certain applications have been made by word embedding[13] or graph-based machine learning,[14] which efficiently encodes materials science data into high-dimensional numeric representations. Nonetheless, this encoding operation makes it difficult to support the outputs with empirical evidence.

An appealing NLP technique called opinion mining is suitable for improving this weakness. It analyses people's emotions, evaluations, and attitudes toward entities such as products and events.[15] Existing opinion mining research mainly relies on lexicon-based and corpus-based methods.[16] The former typically uses a dictionary of sentiment words and phrases, incorporating intensification and negation to compute a sentiment score for each text.[17] The latter usually uses supervised feature-based machine learning or deep learning. Machine learning methods relied on traditional classifiers such

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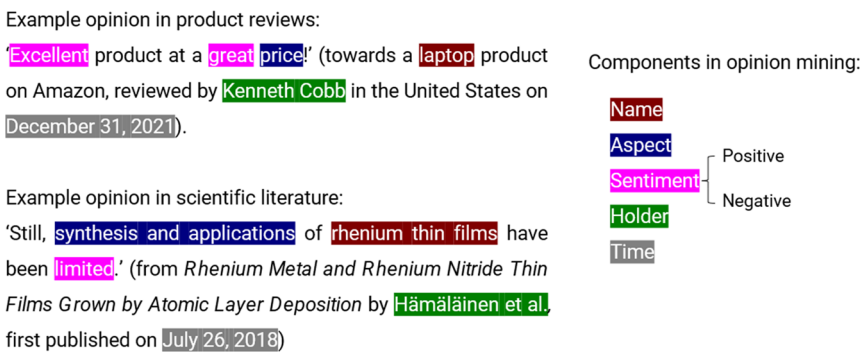


Figure 1. A comparison of opinions in product reviews and the scientific literature. Different highlighting colors are used to indicate various types of components in opinions. In the example of an opinion in a product review, Kenneth Cobb gave a positive sentiment about the price of a laptop on December 31, 2021. In the sample opinion in the scientific literature, a negative sentiment is given about the synthesis and applications of rhenium thin films on July 26, 2018.

as Support Vector Machine (SVM), Naïve Bayes, and logistic regression. With the advent of deep learning, Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and Transformers have been prominent. Opinion mining on Twitter data shows that deep learning methods outperform classical machine learning methods, like Naïve Bayes and support vector machine.[18] Opinion mining has widespread applications: categorizing product reviews to support consumers' decisions,[19] analyzing social media texts to predict the stock market[20] or election results,[21] and fostering interactions between governments and citizens in smart cities.[22]

With the exponential growth of the number of scientific articles, opinion mining has been applied to scientific literature but mostly focuses on citations instead of the article's body content.[23−25] The main purpose of these previous studies is to facilitate researchers to identify research papers they need or analyze the behaviors of authors.[26,27] Most of them construct structural and grammatical features for citation sentence classification. For example, Hassan et al. trained 6 different machine learning classifiers on an annotated corpus to assign the polarity of scientific citations as negative, positive and neutral.[28] Although the content tends to be longer and has fewer opinions, it contains valuable information about research findings. However, related applications have not been reported to date.

Knowledge mining, an efficient and effective way to discover knowledge from texts,[29] can be considered a comprehensive discipline. The knowledge mining methods from texts are usually NLP techniques, including information retrieval (mainly NER), text classification, text summarization, and visualization.[30] One method alone is not enough, and different methods usually work together. For example, in civil engineering, Tian et al. trained a CNN model to classify the construction on-site reports and mined construction knowledge by extracting terms from the classified texts using improved frequency-inverse document frequency (TF-IDF).[31]

In this study, we applied the opinion mining technique to the body content of energy-material-related publications and implemented knowledge mining. For opinion mining, we integrated material knowledge into text representation and constructed opinion data sets to hierarchically train deep learning models, named as Scientific Sentiment Network (SSNet). It is composed of four modules: (i) text preparation, (ii) opinion extraction, (iii) opinion classification, and (iv) information analysis. In module i, we generated a mixed-topic opinion data set from 77 full texts and over 56,000 abstracts (see Methods). Our deep learning models were trained on this data set and achieved high performance in both modules ii and iii, with 94% and 92% accuracy, respectively. Module iv uses mined opinions from publications to mine knowledge using the NER and visualization technique. To illustrate module iv, we designed two material prediction experiments. The first experiment utilized opinions toward thermoelectric materials to improve the rank correlation of experimental results with predicted ranking for maximum power factor from 59%[13] to 70%. In the second experiment, we focus on atomic layer deposition (ALD) elements and their combinations and use a previous distribution of elements to identify the best combinations. For a more refined prediction, we trained the long short-term memory (LSTM) model with 68% accuracy and analyzed the top oxidized candidates. Our findings demonstrated that our system could assess the suitability of materials for an application from various aspects, such as manufacturing performance and natural properties. This work introduces a novel perspective on the assimilation and integration of knowledge, which can provide predictions with evidence to researchers and help them decide on the most promising path forward.

2. OPINION MINING SYSTEM FOR ENERGY MATERIAL SCIENTIFIC LITERATURE

As Liu[15] defined, a regular opinion is a quintuple $(e_i, a_{ij}, s_{ijkl}, h_k, t_l)$ composed of five essential elements: (1) e_i is the name of an entity, (2) a_{ij} is an aspect of e_i, (3) s_{ijkl} is the sentiment on aspect a_{ij} of entity e_i, (4) h_k is the opinion holder, and (5) t_l is the time when the opinion is expressed by h_k. A sentiment s_{ijkl} is usually classified as positive or negative but can also adopt a neutral category and have different intensity levels for each category. The subscripts in the quintuple are used to indicate the correspondence: the sentiment s_{ijkl} is given by opinion holder h_k about aspect a_{ij} of entity e_i at time t_l. Identifying the five components is the essence of opinion mining, because it transforms unstructured text into a structured form that can be further examined.

Similar to typical opinions (such as product reviews), opinions in the scientific literature reflect the advantages or limitations of particular objects. As shown in Figure 1, Hämäläinen et al. gave a sentiment about rhenium thin films.[32] However, product reviews are concise, and each comment can usually be treated as an opinion. In comparison, scientific publications are lengthy texts with subtle expressions. Usually, opinions are hidden and not well expressed, while objective fact and personal bias of authors have to be hedged.

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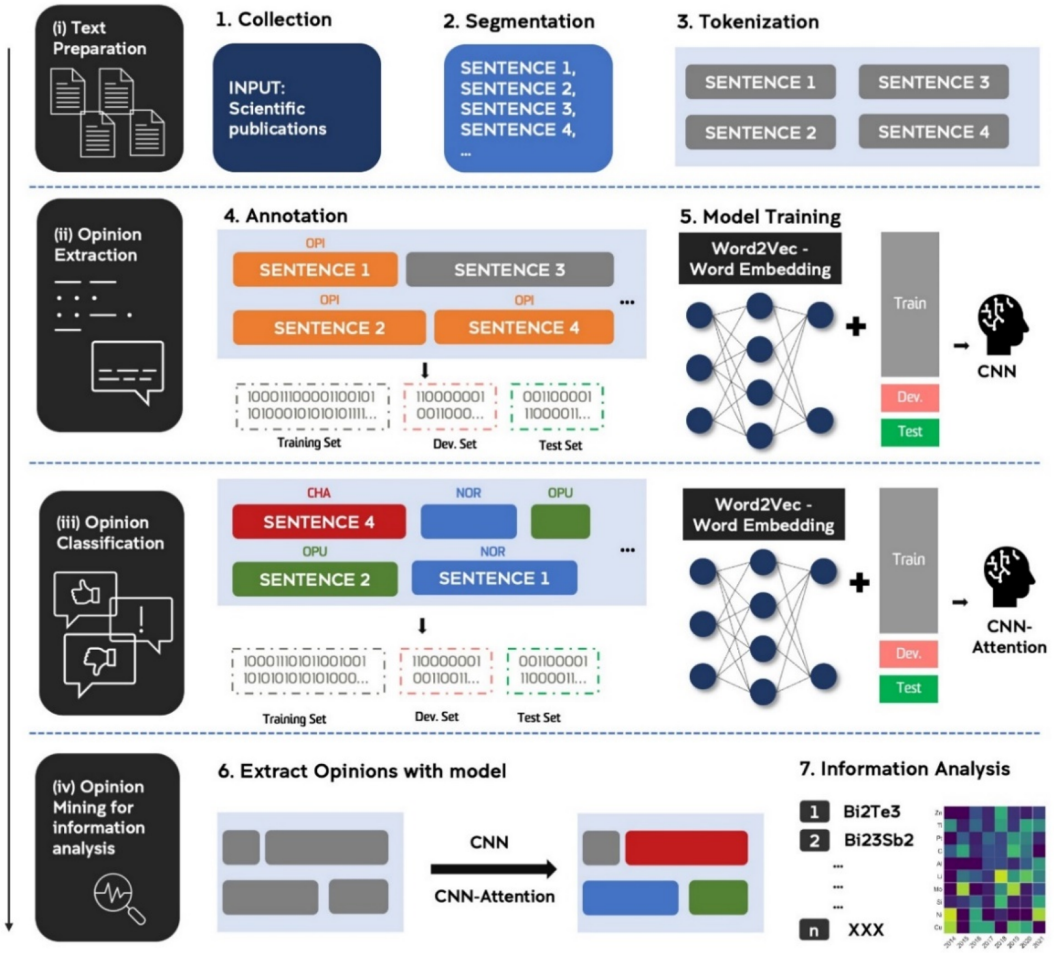


Figure 2. Overview of the training SSNet and the downstream task. The figure illustrates how the four modules work interactively. The first three modules' selected publications were preprocessed and annotated to train models for opinion extraction and classification tasks. In the last module, opinion sentences were extracted from the publications and further classified as opportunities and challenges by trained models for information analysis. During the process, the model's training can be continued with more data to adjust its performance.

We found that opinions about energy materials have a complex and wide range of content, which can be classified as follows: (a) material property; (b) technique, method, and application; (c) experimental finding; (d) social, economic, or environmental implication; and (e) policy and management. Therefore, we adapted the normal process of identifying the five components in the regular opinions. We extracted opinion sentences from publications found by search terms, like "thermoelectric". A search term can be treated as an entity e_i. However, it is difficult to predefine categories of aspects a_{ij} because there are countless materials, methods, and applications. Hence, we moved aspect identification a_{ij} to the last step and proposed using flexible categories for different research purposes. Since our targets are materials, we used normalized element names or material names as aspects in this study. To identify problems and track their progress, we classified the extracted opinions into challenge (negative) and opportunity (positive) categories as sentiment s_{ijkl}. The opinion holder h_k and time t_l were obtained through the metadata of publications, which provided the author's name and year of publication, respectively.

Overall, the opinion sentences were extracted from publications first and then classified as challenges or opportunities to identify their aspects further. This process is implemented in our system as four modules (see Figure 2): (i) text preparation, (ii) opinion extraction, (iii) opinion classification, and (iv) opinion mining for information analysis. The text preparation module i collected publications on a certain topic by searching keywords from databases such as the Web of Science. The publications have different formats, including PDF, JSON, and HTML. Plain texts were extracted from these publications, segmented into sentences, and tokenized (see Methods). The preprocessed sentences were sent to the next two modules. Suppose the text contains the following text:

Lithium transition metal oxides are prevalent cathode materials currently, but they face great challenges due to unsatisfactory energy density, chemical/electrochemical instability, and elemental scarcity concerns.... This synergistic control of nano-/macrostructures is a promising concept for enhancing battery performance and its cycle life. Finally, a rational method of surface modification is proposed for use in cathode materials.

After the module 1−3 process, the text shall be processed and labeled like:

['Lithium', 'transition', 'metal', 'oxides', 'are', 'prevalent', 'cathode', 'materials', 'currently', ',', 'but', 'they', 'face', 'great', 'challenges', 'due', 'to', 'unsatisfactory', 'energy', 'density', ',', 'chemical/electrochemical', 'instability', ',', 'and', 'elemental', 'scarcity', 'concerns', '.'] labelled as "opinion" and "challenge"

['This', 'synergistic', 'control', 'of', 'nano-/macro-structures', 'is', 'a', 'promising', 'concept', 'for', 'enhancing', 'battery', 'performance', 'and', 'its', 'cycle', 'life', '.'] labelled as "opinion" and "opportunity"

['Finally', ',', 'a', 'rational', 'method', 'of', 'surface', 'modification', 'is', 'proposed', 'for', 'use', 'in', 'cathode', 'materials', '.'] labelled as "non-opinion"

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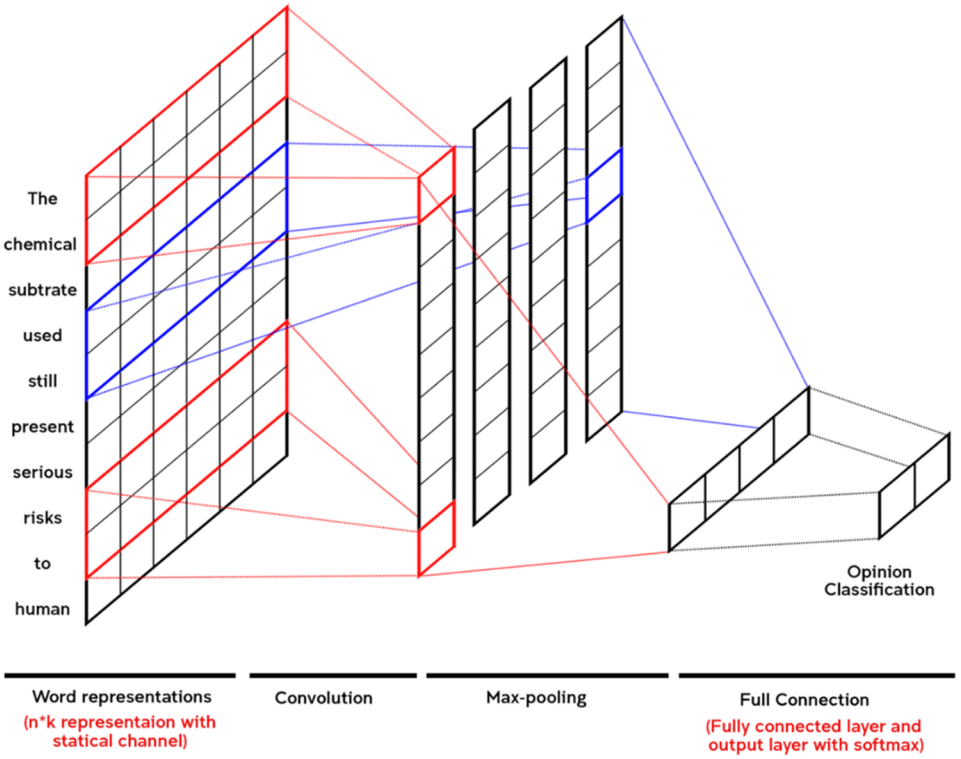


Figure 3. The schematic model architecture of the CNN structure. The CNN architecture is illustrated with an input $n*k$ representation of a sample sentence (the sentence has n-words, and a k-dimensional word vector represents each word). The convolution applied to each sliding window of words (red and blue blocks) generates a feature map. The most significant element from each map is taken by maximum-pooling and fed into the fully connected layer. The result calculated by SoftMax is a regular digit representing "opinion" and "nonopinion" in our task. The detailed working process is available in the Methods.

To choose suitable models for (ii) opinion extraction and (iii) opinion classification, we compared the performance of some typical lexicon-based methods and deep learning models on a small-scale management data set (see Methods and Tables S1−S2 in the Supporting Information). The performance of the convolutional neural network (CNN) and attention-based CNN (ABCNN) model exceeded others in (ii) and (iii), respectively. As shown in Figure 3, the main idea of CNN is that each unit in a layer receives inputs from a set of units located in the neighborhood in the previous layer.[33]

This hierarchical architecture gives CNN high performance on classification tasks, like sentiment classification.[34] The attention mechanism[35] helps the model selectively concentrate on relevant and essential information while ignoring others. Due to the imbalance of opinion and nonopinion categories, we also used Synthetic Minority Oversampling Technique (SMOTE)[36] to improve the performance of the CNN model (see Methods). To construct a larger annotated data set, we used pseudo labels[37] to build a mixed-topic data set with approximately 22,000 samples (see Methods). This larger-scale data set was split into a training set and a test set with a ratio of 19:1 (see Table S3 in the Supporting Information). We trained the final models on this data set. The CNN model for opinion extraction and the ABCNN-SMOTE model for opinion classification achieved 94% and 92% accuracy, respectively. To ensure that the models meet diverse needs, such as retrieving opinions of input sentences at a fixed proportion, we added corpus comparison[38] (see Methods) as an optional function in the opinion extraction stage.

Compared with the prevailing trend of large language models (LLMs) such as BERT[39] and GPT-4,[40] our approach pivots toward a lighter model. We also added MatBERT[41] and EnergyBERT (https://huggingface.co/UNSW-MasterAI/EnergyBERT) for comparison. MatBERT was pretrained on material science literature, and EnergyBERT was continued pretrained on Energy literature (initialized with BERT weights). As Table 1 shows, our model demonstrates performance comparable to, or even superior than, BERT pretrained on energy or material science domain texts, and surpasses the zero-shot performance of GPT-4 (without finetuning). We observed some errors made by each model and found that SSNet, MatBERT and EnergyBERT made less error than BERT on sentences with domain-specific terms and hidden opinions;

Table 1. Performance Comparison between the SSNet with the Large Language Models (EnergyBERT and GPT4) in Opinion Extraction and Classification (See Table S4 in the Supporting Information for a More Comprehensive Comparison)[a]

Model	SSNet	BERT	EnergyBERT	MatBERT	GPT-4
Training Time	352 s	3270 s	3269 s	3273 s	N/A
Inferring Time	<1 s	30 s	30 s	30 s	N/A
Accuracy (Opinion Extraction)	0.9421	0.9267	0.9339	0.9158	0.64
Accuracy (Opinion Classification)	0.9158	0.8173	0.8821	0.8789	0.76

[a]Since GPT-4 does not open for finetuning, we use prompts to predict the label of 50 random examples in the test set for each classification task.

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GPT-4 has a too strict standard for opinion extraction and sometimes predicts opinion as neutral in opinion classification (may caused by label setting of sentiment classification task in instruction tuning of GPT-4); All models tend to make error on complex sentences with contrastive relationship. In addition to high accuracy, SSNet also shows a significantly faster speed for both training and inference, making it well-suited for processing large corpora of scientific literature. This efficiency accelerates the pace of scientific discovery and contributes to the reduction of carbon emissions, aligning with global sustainability goals. Therefore, our methods are designed to integrate seamlessly with LLMs to screen the extracted data. This integration allows us to assign a lower confidence or weight to materials associated with more negative sentiment in the corpus, thereby enhancing the data quality. This improved data, in turn, can guide the design or synthesis of materials more effectively. Detailed cases illustrating the application of our approach are presented in the subsequent sections of this paper, demonstrating the transformative potential of our model in the realm of material science.

In the last module, our system automatically extracted opinions from input scientific texts with trained models for information analysis. For each opportunity or challenge sentence, ChemDataExtractor[3] was used to identify the material name as the aspects of opinion automatically. If no material was recognized, then it returns to the source and extracts a material name nearest to that sentence's position. Each opinion relates to at least one material, and the attitudes toward materials can be analyzed with detailed references and time series. Taking $LaNiO_3$ in experiment 1 as an example, we can analyze the challenges and opportunities in each year in a material-centered view:

```
LaNiO3 {
    '2017': {
        'opportunity':
            ['...Our first-principlebased calculation clearly demonstrates...',
            '...shows that there is a substantial increase in...' ]
        'challenge':
            ['...In contrast to the much studied LaAlO3/SrTiO3 system, the charge
        mismatch at the interfaces...' ]
        }
    '2018': {...}
    ...
    }
```

The following two sections show our design and results of two material prediction experiments as examples of (iv) information analysis.

3. THERMOELECTRICAL MATERIAL DISCOVERY

To determine whether the results from opinion mining correlated with the experimental performance, we generate a rank of thermoelectric materials based on the sentiment ratio and compare it with the rank by experimentally measured maximum thermoelectric power factors (energy conversion efficiency, $z\mathcal{T}$).[42] We used mined opinions from literature related to thermoelectrical materials (see Data and Software Availability) to improve an existing ranking predicted by the Lawrence Berkely National Lab.[13] This ranking prediction of 83 thermoelectric materials is embedding-based (hereinafter referred to as embedding rank) and obtained a 59% rank correlation[43] with experimental results (hereinafter referred to as experiment rank). In contrast, the ranking predicted by a density functional theory (DFT) calculation exhibits only a 31% rank correlation. We collected thermoelectric publications by searching keywords on the Web of Science and extracted their abstracts (see the Methods). After normalization (i.e., Al_2O_3, aluminum oxide, and $Al_{40}O_{60}$ are uniformly replaced by Al_2O_3), 66 materials were found across the embedding rank, the experiment rank, and our collection. For each sentence with any one of these materials, the sentiment was predicted by the trained model. Based on their frequency and sentiment scores, we designed features representing materials' usage rate and sentiment across time (see Note S3 in the Supporting Information).

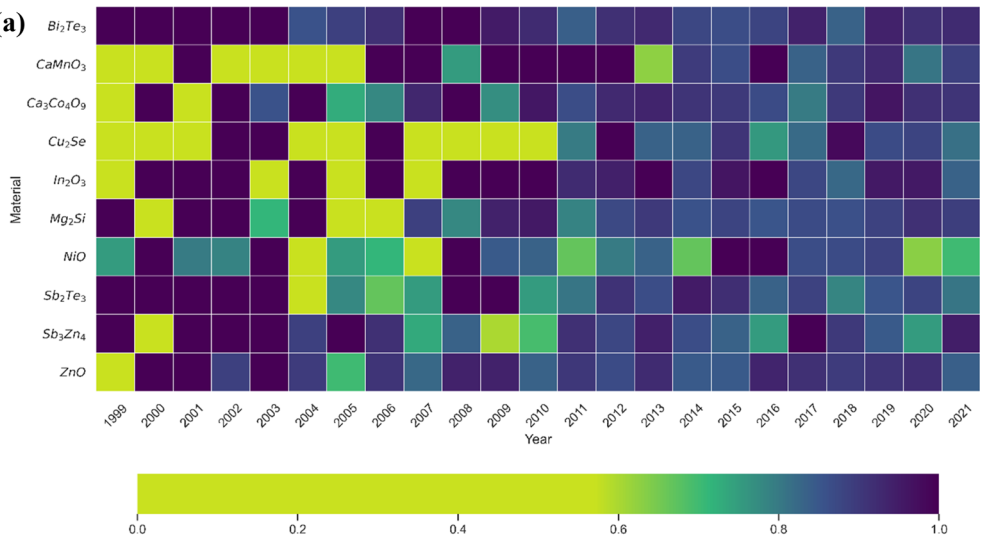
Two alternative approaches were implemented to update the ranking. The first method was to directly modify the embedding rank of 83 materials according to the values of our designed features. The basic idea was to promote the positions of newly emerging materials and lower the positions of the abandoned old materials. The level of change depended on the sentimental value of the material. For example, a newly emerging material with a high sentiment value would be significantly promoted, but the promotion would be less if it has a low sentiment value. With about a quarter of materials' positions being modified by the above rules, the rank correlation with experiment rank was improved from 59% to 69%. The comparison of ranks is available in Note S4 in the Supporting Information.

The second method was based on a supervised learning algorithm and was implemented on 66 materials. We used a pointwise approach to learning the rank algorithm, which takes the feature vector of a sample as the input and predicts the degree of relevance of the sample.[44] We introduced a linear regressor to predict experimental values based on embedding and designed features. The predicted experimental values were used to form the ranking prediction. The average rank correlation between expected ranking and experimental results on 2-fold cross-validation was 70%. The improvement brought by the modification and validation of pointwise ranking showed that our system could reflect experimental facts from the opinions of researchers.

Based on the sentiments of opinions and year of publication, we calculated each material's percentage of opportunities in the opinions by year. This method quantified the sentiment toward materials as sentiment scores across time. Figure 4a visualizes the sentiment scores of the top 10 high-frequency thermoelectric materials across time. Each cell represents the average sentiment value of the element used in thermoelectric materials that year. The sentiment score is represented using cell color; the higher, the darker. Moreover, Figure 4b compares their positions in different ranks. We can see no necessary relationship between the high frequency and high position. For example, NiO is one of the most common compounds in the considered literature, but it only took 79th place in the experiment rank, 82nd in the sentiment rank, and 83rd in the embedding rank. It can be seen in Figure 4a that NiO experienced strong negative opinions. Even though some positive opinions were found in Li-doped,[45] Ca-doped,[46] and Na-doped moieties,[47] researchers have shown concerns over the years, including the fact that NiO-based thermoelectrical materials show a $z\mathcal{T}$ electric material figure of merits compared to other candidates.[48] Moreover, such materials also show poor electrical conductivity, which limits their thermoelectrical application.[49] There were also attempts on nanostructured NiO, which is commonly synthesized for high-performance thermoelectric materials. However, the electrical conductivity of nanostructured NiO appeared to not be significantly higher than normal NiO.[50] After 2010, NiO materials were mainly used as dopants, which helped improve

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(a)



(b)

Top 10 high frequency materials	Embedding Rank	Sentiment Rank	Experiment Rank
1. ZnO	73	69	68
2. Mg_2Si	42	42	15
3. Bi_2Te_3	1	1	1
4. Cu_2Se	25	36	35
5. Ca_3Co_4O9	13	13	14
6. Sb_2Te_3	12	12	9
7. In_2O_3	61	60	55
8. NiO	83	82	79
9. $CaMn0_3$	44	33	47
10. Sb_3Zn_4	26	24	31

Figure 4. Visualization of the sentiment heatmap of top 10 high-frequency thermoelectric materials (1999−2021) and comparison of their ranks by different methods. (a) Sentiment heatmap. The scale from 0 to 1 represents sentiment scores, where 0 indicates a completely negative sentiment or no opinion at all, 0.5 is neutral, and 1 is completely positive. (b) Comparison of the predicted rank of thermoelectric materials by their maximum power factor.

thermoelectrical performance, including in ZnO[51] and $La_3Ni_4O_{10}$.[52] By studying NiO, we can see how sentiment scores and extracted opinions over time provide insights into the suitability of materials for certain applications from different viewpoints.

The results of the perovskite solar cell corpus were utilized to demonstrate the generalizability of our approach (see Methods for details regarding the perovskite solar cell corpus). We used the same SSNet as that for thermoelectric. The sentiment ratio of the elements used in perovskite and carrier-selective materials (both electron selective and hole-selective materials) from 2014 to 2021 is shown in Figure S2−S4 in the Supporting Information.

4. NANOMATERIAL THIN FILM DEPOSITION INSIGHTS

Thin film deposition is a process of depositing a thin layer of material onto a substrate, which is one of the commonly used synthesis methods for nanomaterials. There have already been many previous studies that compare different synthesis methods for the same thin films, for example, sputtering and ALD for ZnO thin films.[53] The ALD process is considered one of the best thin film deposition methods for nanomaterials, since it produces highly uniform and conformal thin films with precise control over thickness. However, the use of ALD is limited by the availability of precursors, the growth rate, and the cost of the process. We reviewed the thin film deposition literature (approximately 600,000 publications from 2000 to 2021) to gain a better understanding of the development of ALD through the opinion mining system.[54] We clustered opinions to the corresponding periodic table elements of materials. Figure 5(b) presents the frequency and sentiment scores of 55 kinds of element combinations in 2014, 2018, and 2021. It is obvious that there are some rising and falling for the sentiments of specific materials. For example, we examined the scientific facts in extracted opinions of Li, an element with obvious changes in the sentiment score. Li experienced a prominent negative sentiment in 2018 due to intrinsic properties, including thermal instability,[55,56] shuttling effects,[57] and electrochemical instability.[58] Fortunately, several improvement measures of these drawbacks were found in opinions in the following years. In 2019, Li provided the possible solution for the $LiMn_{1.5}Ni_{0.4}O_4$ (LMNO) nanomaterial cycle time through selectively con-

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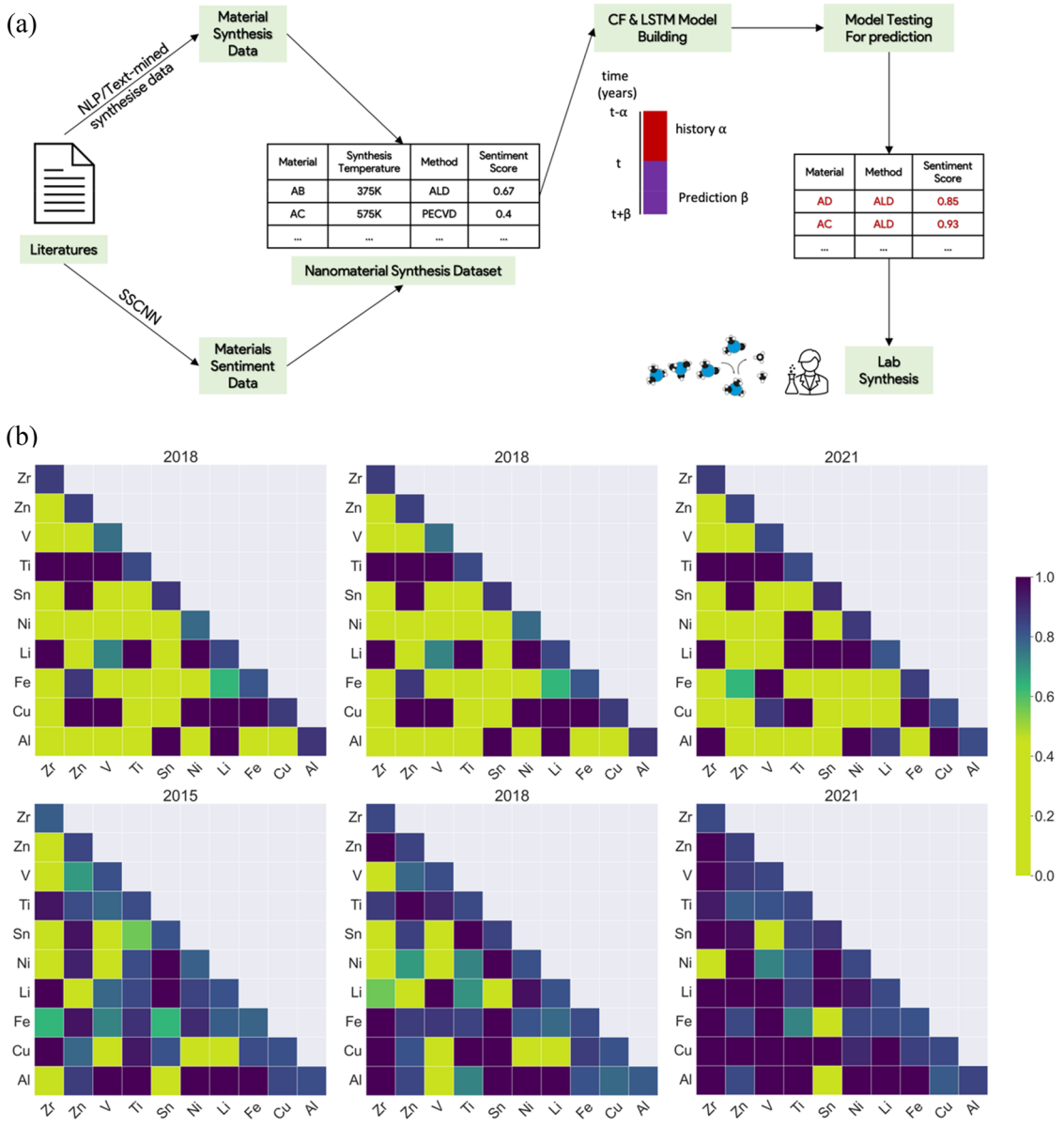


Figure 5. (a) The system architecture of nanomaterial synthesis methods prediction using collaborative filtering (CF) and long−short-term memory (LSTM) model. (Parameters of the LSTM model: length of the history data α, response time β.) The detailed working process is available in the Methods. (b) Sentiment heatmaps of elements used in ALD (top 3) and other deposition methods (bottom 3) of 2014, 2018, and 2021. Since scientific sentences typically emphasize positive relationships, we reduce the span of light colors and set the most negative sentiment as 0.

trolled methods, integrating ALD with annealing at the nanolevel and process controlling and optimization at the macro level.[59] After that, Lou et al.[60] showed that the addition of Ti to Li-compounds can improve its cycling stability which can easily be achieved using ALD. The lithium case demonstrates our system's competence to provide key insights from challenges and opportunities, inspiring researchers to explore potential solutions in interdisciplinary scenarios to research questions.

To allow quantitative prediction, we enlarged the data set to thin-film deposition-related full-text papers (18.2 M sentences) and divided them into ALD and non-ALD corpus. SSNet labeled materials on extracted and classified opinion sentences. We clustered materials as 2- or 3-element combinations (i.e., Ti−O, Ti−Al−N) and classified them as N, O, and S series. The statistics show that about 79% of ALD element combinations were from non-ALD fields, and the time gap between one material first deposited as a thin film and ALD deposited thin film is from 2 to 9 years. Based on this observation, non-ALD element combinations which have not appeared in the ALD data set should be candidates with good potential.

We tested two machine learning methods to predict candidate element combinations. The first method draws on the transition probabilities of the elements through collaborative filtering (CF). CF is a classical method in recommendation systems, and its main idea is similar users will like similar items. In our scenario, we treated element combinations successfully transformed from non-ALD deposition methods to ALD for a period similar to similar combinations. The distribution of these combinations is calculated as element scores. For each year, we ranked the current non-ALD candidates unused in ALD by their average element score. For instance, in 2013, O−Fe−Ti was ranked the highest among 924 non-ALD candidates in the O series and was found to appear in 2015. On average, the top 5 successful predictions (see Note S5 in the Supporting

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Information) were ranked top 3% in hundreds of candidates, and we display predictions for 2022 for each series (see Note S6 in the Supporting Information).

To utilize temporal information, the second method involves training an LSTM model. LSTMs are recurrent neural networks commonly used for predicting and classifying sequence data. As shown in Figure 5a, an LSTM model can be trained in the context of nanomaterial synthesis methods based on a data set of known synthesis methods and their corresponding properties, such as target materials, synthesis temperature, and methods, along with average sentiment scores. After training, the LSTM model could be used to predict synthesis methods given a set of input properties.

We analyzed whether these models could accurately predict the materials reported in the literature in recent years when trained on data available in the past. Specifically, we divided the sentiment corpus into two parts, the training data set (from 2000 to 2020) and the test data set (2021). The CF and LSTM models were trained separately to generate the possibility that the oxidants could be reported in 2021 (test year). To ensure that there are adequate training samples for the LSTM model, we selected 15 of the most popular elements that have been grown by ALD. We calculated the average accuracy of each prediction result. 95 types of oxide were grown using ALD by 2021, and the LSTM model predicts 58 of them, while the CF model predicts 34 of them. Based on the two models, Table 2 shows the top ten results for the prediction that oxides could be synthesized through ALD after 2021. Three results appear in both rankings. The named entities were identified and classified based on their properties, application methods, and deposition modes using NER. We have conducted an extensive manual literature search in order to find predictions for which we could find any ALD reports outside of our corpus. The mainstream manufacturing method for these oxides is sputtering. It has been proved that ALD coating has superior coating integrity in TiO_2 films than sputtering coating.[82] Similarly, we also expect that the oxides in Figure 5 will obtain better performance with ALD in the future, including optical, thermal, electricity, and stability. These materials are mainly applied in random access memory (RAM), sensors, and photovoltaics.

Theoretically, the LSTM model should be able to capture long-term dependences in the data. Furthermore, it can retain information about past material synthesis results relevant to current problems. Finally, the complex relationships between different synthesis methods and their corresponding material and natural language features. These features allow the LSTM model to make more accurate predictions compared to CF.

Table 2. List of Top 10 Oxide CF and LSTM Predictions Based on the Opinion Corpus at the Time of Writing

Oxides	Properties	Application	Deposition Methods
Al, V	Electrochemical, carrier selectivity	Semiconductor, sensor	Sol−gel, spin coating[61]
	High dielectric constant	Dielectrics	
Al, Bi	Photocatalysis	Water splitting	Spray pyrolysis, spin coating, sol−gel[62−65]
	Thermoelectric	Energy storage	
	Ferroelectric	RAM	
Ga, Ti	Optoelectronic photocatalytic	Solar cells	Sputtering,[66,67] solid states reaction[68,69]
		Photocatalysis	
		Chemical sensors	
Al, W	Photocatalytic spectral selectivity	Selective absorbing coatings	PVD,[70] sputtering[71,72]
	thermal stability	Energy storage	
Fe, Ga	Magneto-optical	Solar cells	PLD,[73] sputtering,[74] electrodeposition[75]
	Magnetoelectricity	Gas sensors	
		Lateral field effect transistor	
La, P[a]	Carrier Selectivity	Solar cell	Wet Chemistry,[76] Spin Coating,[77] Mechanical milling,[78] Sol−gel[79]
	Photoluminescent	Sensor	
		Calcined catalysts	
Co, Cu	Photocatalysis	Catalysts, sensors, solar cells,	Spin Coating,[80] Hydrothermal[81]
	Transparent		
	Conductive		

[a]Note: Oxides reported that used ALD to synthesize them in a Wiley publication.

5. CONCLUSION

Opinions in materials publications contain valuable information about energy materials. We adopted the theory of regular opinion mining in scientific texts and implemented a system that can extract and classify opinions in the energy material literature. It was designed for the main body of scientific publications rather than citations or review texts, providing sentence-level information about the authors' attitudes toward energy materials. Each mined opinion can be traced to the original article with provenance instead of giving a final rank or score, which helps to integrate cross-disciplinary information into quantitative analysis. During the training, we alleviated the burden of annotation by data augmentation to such a level that individual experts could handle it, addressing the general concern that deep learning requires vast amounts of annotated data. In material prediction experiments, we apply an NER tool to utilize small information units such as material names. Our prediction by an LSTM model generates dependable insights which oxides (e.g., Al_xV_yO) have the potential to be synthesized by ALD. These oxides are mainly applied in RAM, sensors and photovoltaics fields. Compared with similar text classification tasks on material publications,[11,83,84] our system demonstrates a more straightforward end-to-end framework with a more robust semantic competence and greater interpretability.

In this study, some training data were from abstracts of publications. On the one hand, using abstracts allows extensive coverage of various research areas. On the other hand, abstracts are more likely to include general named entities, such as "perovskite" and "photovoltaics", and emphasize positive results. An opinion sentence might occasionally have more than one named entity with different corresponding attitudes, causing erroneous opinion classification. Our future work involves the use of a pretrained language model with more energy knowledge and finer-grained opinion mining with more specific material names. Energy knowledge can be integrated into the model by using word embedding pretrained on energy text for input representation or concatenating with energy entity embedding. Current opinion mining is on the sentence level, but we observe that there sometimes exist more than one material name and corresponding different opinions. A finer-grained opinion mining may be achieved by preprocess of subsentences with multiopinion expression trigger detected. For improvement of the accuracy of the nanomaterial thin film insights machine learning model, additional related parameters could be the available reactant groups for specific elements (e.g., Halides, Cyclopentadienyls) and different kinds of counter reactants

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(e.g., H_2O, O_3, NH_3, H_2S). Moreover, we look forward to extending our system to more applications relying on or related to opinion mining in scientific texts, such as real-time nanotechnologies opinion search engines.

6. METHODS

Data Collection and Preprocessing. We first built a small-scale management data set with the End-of-Life (EoL) topic, which is rather rich in opinions. The full text of 77 highly cited publications was used to train the raw model. In preprocessing of EoL full papers, the references and other publication information were removed using regular expressions in preprocessing. After text cleaning, the full text was divided into sentences. We designed a sentence tokenizer for publications based on NLTK sentence tokenizer with high accuracy on some expressions, like "et al.", "Figure", and "e.g." (these expressions are easier to wrongly justify as the end of sentences in existing sentence tokenizers). Two PhD candidates who majored in materials science annotated the EoL data set (see Note S1 in the Supporting Information), and we also invited the authors of some publications in the test set to correct the annotations.

We then prepared a mixed-topic data set by adding samples from material and manufacturing data set to EoL data set. To make sure the balance of mixed-topic data set, we use three different kinds of topics: End-of-Life (management), perovskites (material), and atomic layer deposition (manufacturing method). They are in a parallel relationship. Using keywords search (use "perovskite" and "ALD"), the publications were retrieved from Elsevier's Scopus and Science Direct application programming interfaces (APIs), and the records were saved as text files containing all meta fields provided by Web of Science. A two-character field tag is prefaced in each area. For example, the abstract text was prefaced with "AB". The key information in plain text files was converted to the structured dictionary form. The trailing copyright information in the abstracts was removed, and the abstracts were divided into sentences with the sentence tokenizer mentioned above. The raw model predicted the annotation of sentences in these two data sets to form pseudo labels. These labels were assigned automatically by trained raw models: this kind of method cannot ensure annotation quality but avoids the time-consuming large amount of human annotation. We split the mixed-topics data set into train/dev/test sets and proofread the opinions labels to guarantee the final metrics' authenticity.

As for thermoelectrics, the data are collected in a similar way as perovskite and ALD. It does not belong to any topic in the mixed-topic data set, and it was not used in training models.

As for thin film, a self-built paper downloading tool (https://github.com/MasterAI-EAM/SciCrawler) was used to collect scholarly literature returned by WoS search query. We only used related full-text papers published by Elsevier, Springer Nature or Royal Chemical Society and converted them into plain texts (about 18.2 M sentences). The texts were divided into ALD and non-ALD types by keywords. Since ALD is one of thin film techniques, we also added sentences in the ALD data set (ALD topic in mixed-topic data set) into the thin film data set if it was not seen in this data set. Thin film covers ALD, so we call it as "enlarge".

Lexicon-Based Approaches. TextBlob is a Python library actively used in sentiment analysis research about product reviews and social media texts. We built a hierarchical classifier based on the TextBlob lexicon-based sentiment analyzer. The analyzer returns the polarity and subjectivity for each input sentence. Polarity is in $[-1, 1]$, indicating the sentiment trend (0 represents neutrality, and a larger value represents more positivity). Subjectivity is in $[0, 1]$ and the larger it is, the more likely the sentence contains opinion, emotion, or judgment. These numerical values are computed based on a precompleted dictionary with one or more entries of each word. TextBlob uses a sophisticated averaging technique to calculate the partition for a single word. Therefore, a similar operation for each term in longer texts would give a combined section. We considered both properties, and the results were slightly better than those of the Naïve Bayes model, achieving 64% and 75% accuracy in two stages. The metrics indicate that opinion mining in the commercial field is not entirely suited to academic papers, and we manually created a sentiment dictionary. We counted the word frequency statistics of content words (nouns, adjectives, verbs, and adverbs) and manually selected opinion words from high-frequency content words. High-frequency words imply that they are more representative and can influence more sentences in classification. Content words are more meaningful than function words. We aim to achieve more accurate sentiment score calculations with a limited number of words selected. We also observed that modal verbs (can, could, should, and so on) play an essential role in opinion sentences and score higher for their appearance. With the sentiment dictionary and the point scale, we built a point system to calculate the score of every sentence. Each publication put into the point system will return the selected sentences according to demand, for example, the top 20%, as the opinion sentences of this publication. Based on this domain-specific sentiment dictionary, the accuracy of the opinion extraction is improved to 72%, but the accuracy of opinion classification declines. More details are available in the Note S2 in the Supporting Information.

CNN-based Models. The working process of CNN can be summarized as follows:

(1) Word representations: A vector represents each word in the sentence. Thus, textual sentences can be converted to numeric inputs.

(2) Convolution: Different filters are used on the word vectors to create convolutional feature maps. Filters are weight matrices of predefined size (red and blue squares in Figure 2), which slide horizontally across the sentence. A convolutional feature map is formed by summing the component-wise multiplication of data and corresponding filters' weights at each step. After the filter "scan" of the input sentence, a convolutional layer comprises a series of those convolutional feature maps.

(3) Max-pooling: This operation takes the most significant element from each map to learn high-order representation and reduce dimensions simultaneously.

(4) Full connection: the matrices are flattened into the vector and fed into the fully connected layer. An activation function in the output layer, such as SoftMax, classifies the outputs as regular digits. The backpropagation algorithm adjusts the network weights to minimize the cost function.

The CNN model we used in opinion extraction has one layer of convolution and one layer of max-pooling. Compared with the extraction task, two more convolutional layers and an attention mechanism at the maximum-pooling step were implemented in the CNN-Attention model for opinion classification. There are three components in the attention mechanism: the queries Q, the keys K, and the values V. The

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attention score, a mask matrix, is calculated by the SoftMax dot product of Q and K. The attention score can weigh the original sequence represented by V. In the CNN-Attention architecture we used, the attention score is calculated by the outputs of convolutional layers. The attention score and initial results of convolutional layers are concatenated after max-pooling on both and fed into the fully connected layer.

We implemented both the CNN model and the ABCNN model with an NLP transfer learning framework called Kashgari (https://github.com/BrikerMan/Kashgari). The word representations we used were publicly available 200-dimensional word embeddings (https://github.com/materialsintelligence/mat2vec) trained on 3.27 million materials science abstracts.[13] As for the model's hyperparameters, we tried some combinations and selected the one with the highest accuracy on the development set: dropout of 0.4, batch size of 64, and epochs = 30. Since the number of barrier sentences is much smaller than driver sentences, we also used an augmentation method, SMOTE, to balance the categories in opinion classification. This method replaced content words in the challenge sentences with their most similar words found by the cosine similarity of word vector and used these sentences to enrich the data.

Data Augmentation. We used SMOTE and pseudo labels to generate more data for training. In the first method, we replace some content words in the training samples with similar words. The size of samples in the minority class is doubled through this way. In the second method, the CNN-SMOTE and ABCNN model trained on the EoL data set were used as raw models to predict the pseudolabels of sentences in abstracts of ALD- and perovskite-related publications. Annotators were invited to proofread the predictions, and over 90% of predicted opinions were true, proving that the models have good scalability in similar fields. After the proofreading, the annotated sentences in three different topics were merged to build a mixed-topic data set with approximately 22,000 samples. The use of data augmentation significantly reduced the amount of manual labor required.

Corpus Comparison. Though the deep learning method shows significant advantages in opinion extraction, it can only predict categories instead of measuring the degree of opinion-like. We implemented a function called corpus comparison to extract the top x% most opinion-like sentences from the input based on the results of the CNN model. We first calculated the termhood of words, and the words were sorted into descending order. The top candidates formed a dictionary, and we used the same points system in the lexicon approach to calculate the opinion-like scores of sentences. If the number of extracted sentences cannot fulfill x% requirement, most opinion-like sentences in the rest will be selected. If the number of extracted sentences surpasses the x% requirement, the extra least-opinion-like sentences will be removed.

The basic idea of calculating termhood is that if a word has a higher significance in the domain corpus and a lower significance in the background, it should be given a greater termhood. In this scenario, we treated the collection of nonopinion sentences as background corpus and the collection of opinion sentences as domain corpus. We experimented with two different termhood measurements: rank difference[38] and frequency ratio.[85] Given a domain corpus D (with a vocabulary V_D) to represent a subject field and a balanced corpus B (with a vocabulary V_B) as background, the termhood of a candidate word w calculated by rank difference was defined as

$$\tau(w) = \frac{r_D(w)}{\sum_{w' \in V_D} r_D(w')} - \frac{r_B(w)}{\sum_{w' \in V_B} r_B(w')}$$

Note: $r(w)$ is w's rank by word frequency in a corpus.

With the same setting, the term hood of a candidate word w calculated by frequency ratio difference is defined as

$$\varphi(w) = \frac{f_D(w)}{\sum_{w' \in V_D} f_D(w')} - \frac{f_B(w)}{\sum_{w' \in V_B} f_B(w')}$$

Note: $f(w)$ is w's frequency ratio in a corpus.

In our experiment, the second termhood measurement had better performance. We also tried different numbers of candidates to form dictionaries. However, there is no real distinction between these versions and those with fewer candidates. Thus, we finally used the frequency ratio difference to calculate termhood and used the top 200 candidates to form the dictionary. Thus, we finally used frequency ratio difference to calculate termhood and used the top 200 candidates to form the dictionary.

LSTM Models. Table 3 illustrates the features extracted from the thin-film deposition opinion corpus. The LSTM process is conditioned on two random variables, namely, history window size and prediction window (as shown in Figure 5a). Our prediction model uses features extracted from the history window and predicts the sentiment changes in a future window.

Table 3. Time Series Features Extracted from the Thin-Film Deposition Corpus to Model the Development of Deposition Methods

Feature Type	Description
Atomic Descriptor	Number of electrons in sharp, principal diffuse and fundamental orbitals
Sentiment Ratio	#Positive: #negative opinion
	#Postive: #all opinions
	#Negative: #all opinions
Synthesis Condition	Average thin film deposition temperature
Synthesis Methods	ALD, others

History windows indicate the length of the history data sequences trained for prediction. If t = 2021 and α = 1, then the data from 2020 to 2021 were used for prediction. Prediction windows are designed to adjust the smoothing process, which is used to extract the noise from the signal. As a result of sentiment changes, there is a substantial amount of noise associated with them. Due to the long time required for experiments to be carried out from the point of conception to publication, especially for some low-frequency materials, there is a considerable amount of noise. If the beta is set at 3 years, we will calculate sentiment changes over the next 3 years instead of using a 1-year interval. Thus, when training our machine learning model, we can avoid overfitting. To determine the optimal set of parameters, we experimented with several combinations of parameters. The best size for the history window is 8 years, and the best size for the prediction window is 1 year.

■ ASSOCIATED CONTENT

Data Availability Statement

The main codes of methods mentioned in the study are available at https://github.com/MasterAI-EAM/TextMaster. Data of

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each topic in the mixed-topic data set can be found in this Github repository (need to download zip files). The academic literature included in this study comprised relevant papers published in scientific journals, conference papers, books, and book chapters. Government and industry reports and other gray documents were excluded due to difficulties in systematically collecting. The search queries and conditions are shown as follows. Search query of EoL publications (on Science Direct): "end of life" OR "recycl*" OR "waste management" AND "photovoltaics" OR "solar panel". Search query of perovskite publications (on WoS): "perovskite" and "solar cell". Search query of ALD publications (on WoS): "atomic layer deposition". Search query of thermoelectric publications (on WoS): "thermoelectric" OR "thermoelectrical" OR "thermoelectric" OR "thermoelec*" Search query of thin-film (on WoS): "thin film". Period for all search queries: 2000 to 2021 (01-06-2021). Language: English.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jcim.3c00746.

> Data splitting of EoL data set and mixed-topic data set, definitions of category in training data, comparison of methods in implementing raw models, evaluation details of raw models and final models; feature design of ranking prediction, complete comparison of rank prediction, successful prediction of element combinations from 2011 to 2021 and prediction of 2022, proportion of publication category; visualization of most frequent elements and most frequent materials; and visualization of sentiment values (PDF)

AUTHOR INFORMATION

Corresponding Authors

Tong Xie − *School of Photovoltaic and Renewable Energy Engineering, University of New South Wales, Kensington, NSW 2052, Australia; GreenDynamics Pty. Ltd., Kensington, NSW 2052, Australia;* orcid.org/0000-0002-1659-4865; Email: tong.xie@unsw.edu.au

Chunyu Kit − *Department of Linguistics and Translation, City University of Hong Kong, Kowloon Tong, Hong Kong;* Email: ctckit@cityu.edu.hk

Bram Hoex − *School of Photovoltaic and Renewable Energy Engineering, University of New South Wales, Kensington, NSW 2052, Australia;* orcid.org/0000-0002-2723-5286; Email: b.hoex@unsw.edu.au

Authors

Yuwei Wan − *Department of Linguistics and Translation, City University of Hong Kong, Kowloon Tong, Hong Kong; GreenDynamics Pty. Ltd., Kensington, NSW 2052, Australia*

Haoran Wang − *School of Photovoltaic and Renewable Energy Engineering, University of New South Wales, Kensington, NSW 2052, Australia*

Ina Østrøm − *School of Photovoltaic and Renewable Energy Engineering, University of New South Wales, Kensington, NSW 2052, Australia;* orcid.org/0000-0001-7159-2574

Shaozhou Wang − *School of Photovoltaic and Renewable Energy Engineering, University of New South Wales, Kensington, NSW 2052, Australia; GreenDynamics Pty. Ltd., Kensington, NSW 2052, Australia;* orcid.org/0000-0002-7299-3481

Mingrui He − *School of Photovoltaic and Renewable Energy Engineering, University of New South Wales, Kensington, NSW 2052, Australia*

Rong Deng − *School of Photovoltaic and Renewable Energy Engineering, University of New South Wales, Kensington, NSW 2052, Australia;* orcid.org/0000-0002-1452-8020

Xinyuan Wu − *School of Photovoltaic and Renewable Energy Engineering, University of New South Wales, Kensington, NSW 2052, Australia*

Clara Grazian − *DARE ARC Training Centre in Data Analytics for Resources and Environments, South Eveleigh, NSW 2015, Australia; School of Mathematics and Statistics, University of Sydney, Camperdown, NSW 2006, Australia*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.jcim.3c00746

Author Contributions

¶T.X. and Y.W. contributed equally. T.X. and Y.W. conceived of original research idea and wrote the codes of system and visualization. T.X., H.W., S.W., M.H., R.D., and X.W. performed data collection, annotation and management. T.X. interpreted the results of information analysis. All authors edited and reviewed the manuscript. C.G., K.C., and B.H. supervised the work.

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Notes

The authors declare no competing financial interest.

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